NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

AVR MCUs

Ideal for sensor applications, ATtiny3217 and ATtiny3216 microcontrollers (MCUs) bring the benefit of two analogue-to-digital converters (ADCs) that enable systems to implement touch control simultaneously with other analogue measurements. One ADC can be used with peripheral touch controller for touch signal acquisition, while the second for monitoring inputs such as thermistors and pressure sensors. Or, both ADCs can be used for faster sampling of different types of sensors.

Additional benefits include:

1. Improved real-time performance and accuracy
2. Robust and reliable performance with built-in safety functions
3. Improved noise immunity and functionality in extreme environments, as devices operate at up to 5V and come in 125°C variants
4. Increased functionality with more room for application code with 32kB flash

Microchip Technology Inc.
www.microchip.com

DC-to-DC converter

Traco Power releases THM 15 family of medical 15W DC-to-DC converters with a wide 2:1 input range. THM 15 series are a reliable solution not only for medical equipment, but also for demanding ITE. The converters provide high isolation of 5000V_{AC} RMS (250V_{AC} RMS working voltage) with reinforced insulation and low leakage.

Traco Power
www.tracopower.com

Step-down switching converters

EMI-compliant, Himalaya step-down switching converters from Maxim Integrated significantly reduce design cycles and enable quick time-to-market. Compliant with CISPR 22 and EN 55022, these are ideal for general-purpose applications, including industrial, building automation, factory automation, communications and consumer electronics.

The regulators make use of latest advancements in switching regulator and process technology, to minimise power dissipation, temperature rise, solution size and total cost-of-ownership.

Maxim Integrated Products Inc.
www.maximintegrated.com

RF module

Ideal for a wide range of applications, Digi XBee3 ZigBee 3.0 embedded radio

frequency (RF) modules are pre-certified for secure and reliable connectivity in a simple, compact micro form factor. These can be easily configured—locally or remotely—using simplified AT or API commands, Digi XCTU software or Digi Remote Manager.

Some features are:

1. Out-of-the-box RF communications; no configuration needed
2. Point-to-multipoint network topology
3. 2.4GHz for worldwide deployment
4. Common Digi XBee footprint for a variety of RF modules
5. Sleep current of sub-1μA
6. Firmware upgrades via UART, SPI or over-the-air (OTA)
7. Migrate to DigiMesh and 802.15.4 protocols and vice-versa

Harihi Ohm Electronics
www.harihiohm.in

Controllers

Texas Instruments' (TI's) DLPC347x controllers offer micron-to-sub-millimetre resolution typically found in high-performance, industrial-grade applications in a smaller form factor for desktop 3D printers and portable 3D scanners.

Developers can pair DLPC3470, DLPC3478 or DLPC3479 controller with one of the four existing DLP Pico digital micromirror devices (DMDs) to design a variety of battery-powered and handheld designs. High resolution and fast switching speed of the DMDs and new controllers enable 3D printing speeds five times faster than existing technologies. With precision pattern control in a smaller form factor, developers can capture fine details of an object with micron-level accuracy, which benefits applications like dental scanning, 3D modelling and 3D vision for robotics.

Texas Instruments
www.ti.com

Controller IC

ICL5102 by Infineon Technologies integrates power factor correction (PFC) and half-bridge controllers in a single DSO-16 package. It supports universal input voltages ranging from 70V AC to 325V AC, and has a comparable wide output range. A low number of external components are required to configure and support this controller integrated circuit (IC). All parameters are set by resistors.

ICL5012 supports fast startup under 500ms at less than 100µA. PFC is greater than 99 per cent, and total harmonic distortion is less than 3.5 per cent. The controller has up to 94 per cent efficiency by resonant topology. Active burst mode for low standby consumes less than 300mW. An enable/disable function supports dimming.

Infineon Technologies India Pvt Ltd
www.infineon.com